Determination of Chemical Reaction Order Using the Box–Cox Transformation

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Abstract

Determining reaction order is a central task in chemical kinetics. In Physical Chemistry B, conventional approaches often rely on trial-and-error linear regression or the half-life method. The former is constrained by the available computational tools and can be difficult to apply to noninteger reaction orders, whereas the latter places substantial demands on the distribution of data points and the timing of experimental sampling.

This paper proposes an approach to determining reaction order based on statistical principles. The Box–Cox transformation is applied to the original time–concentration data, and an optimal transformation parameter is identified to linearize the kinetic equation and thereby determine the reaction order. This approach has the potential to overcome the limitations of conventional methods in identifying half-integer orders and to improve the general applicability and accuracy of interpreting time–concentration or time–partial-pressure curves.

Keywords: chemical reaction rate; reaction order; Box–Cox transformation; maximum likelihood estimation; linear regression

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1. Introduction

Determining the rate equation of a reaction is important for both engineering design and scientific research, and determining the reaction order is a key step in establishing that equation. Reaction order must be determined experimentally. By monitoring changes in the concentration of a substance during a reaction and appropriately processing the resulting concentration–time data, the reaction order can be obtained.

The following methods can be used to determine the order of a chemical reaction: [1]

Trial-and-error method. The rate equations for zero-, first-, and second-order reactions can be written, respectively, as:

$$k = \frac{C_0 - C_A}{t}$$

$$k = \frac{1}{t} \ln \frac{C_0}{C_A}$$

$$k = \frac{1}{t} \left(\frac{1}{C_A} - \frac{1}{C_0} \right)$$

Half-life method. Using the equation

$$\log_{10} t_{1/2} = (1 - n) \log_{10} C_0 + \log_{10} A$$

the reaction order can be obtained from the slope of the straight line:

$$n = 1 - \text{slope}$$

However, both methods have clear limitations. Trial-and-error linear regression is essentially a multiple-choice exercise: it selects the closest fit from a set of predetermined integer orders. Many real complex reactions, such as certain chain or catalytic reactions, may have orders of 0.5 or 1.5. Because calculators cannot automatically search for the optimal exponent, students dealing with a noninteger order often have to settle for an integer order with a relatively high correlation coefficient. The resulting assignment may therefore deviate from the true mechanism.

Although the half-life method can theoretically be used to calculate any reaction order, applying it to time–concentration problems or experimental data requires accurately knowing the times at which the concentration falls to particular fractions of its initial value, such as 50% or 25%. In experiments, or in data supplied in examination questions, measurements are often taken at fixed intervals—for example, every 10 seconds. It is therefore difficult to obtain a measurement at precisely the moment when the concentration halves. The time often has to be estimated by plotting the data and drawing a trend line by hand, introducing substantial operator-dependent error.

It is worth noting, however, that the mathematical transformations applied to different reaction orders in the trial-and-error method are conceptually consistent with transformations used in statistics. The idea of seeking a linear relationship by changing an exponent is closely related to the Box–Cox transformation, a standard statistical tool for handling nonlinear data.

Proposed by George Box and David Cox in 1964, the Box–Cox transformation is a power transformation primarily used to make non-normally distributed data more nearly normal or to linearize nonlinear relationships. Its standard mathematical form is: [2]

$$y^{(\lambda)} = \begin{cases} \frac{y^{\lambda}-1}{\lambda}, & \lambda \neq 0, \\ \ln y, & \lambda = 0. \end{cases}$$

In chemical kinetics, the observed concentration C, or partial pressure P, can be used as the variable to be transformed. By adjusting the transformation parameter λ, this formula covers nearly all the linearizing transformations commonly used in kinetics:

- When $\lambda = 1$, it corresponds to the linear form of a zero-order reaction.
- When $\lambda = 0$, it corresponds to the logarithmic form of a first-order reaction.
- When $\lambda = -1$, it corresponds to the reciprocal form of a second-order reaction.

The Box–Cox transformation therefore encompasses the mathematical transformations needed to identify common zero-, first-, and second-order reactions. Its greater advantage is that λ can take continuous values. This means that the method can also identify reactions with less conventional orders, such as complex kinetic systems with orders of 0.5 or 1.5. Compared with manual trial and error using discrete candidate orders, it provides a more general framework capable of handling arbitrary reaction orders. This paper therefore adopts the transformation as the basis for an improved method of determining reaction order.

2. Results

2.1 Relationship Between the Box–Cox Parameter λ and Reaction Order n

To establish the mathematical relationship between the Box–Cox parameter λ and reaction order n, we begin with the integrated form of the kinetic rate equation and compare it with the definition of the Box–Cox transformation.

Suppose the rate equation is:

$$-\frac{dC}{dt} = kC^n$$

Integration must be considered in two cases.

Case A: $n \neq 1$. Separating the variables gives:

$$\int_{C_0}^{C} C^{-n}\, dC = \int_{0}^{t} -k\, dt$$

Integration yields:

$$\frac{C^{1-n} - C_0^{1-n}}{1-n} = -kt$$

This can be expressed as a function of C:

$$f(C) = \frac{C^{1-n}}{1-n} = -kt + A$$

Case B: $n = 1$. Separating the variables gives:

$$\int_{C_0}^{C} \frac{1}{C}\, dC = \int_{0}^{t} -k\, dt$$

Integration yields:

$$\ln C - \ln C_0 = -kt$$

or:

$$\ln C = -kt + \ln C_0$$

For a non-first-order reaction, set $y = C$ and $\lambda = 1 - n$ in the standard Box–Cox expression. The transformation then becomes:

$$\frac{C^{1-n} - 1}{1-n}$$

This is mathematically equivalent to the integrated kinetic equation

$$\frac{C^{1-n} - C_0^{1-n}}{1-n} = -kt$$

in the relevant respect: the denominators are identical, and the difference between the constants in the numerators affects only the intercept, not the linear relationship between the transformed concentration and time.

For a first-order reaction, the relationship $\lambda = 1 - n$ gives $\lambda = 0$. The Box–Cox transformation is then defined as $\ln y$, or $\ln C$, exactly matching the integrated first-order rate equation.

Table 2. Reaction order, kinetic form, and corresponding Box–Cox parameter

- $n = 0$: t versus C; $\lambda = 1$
- $n = 0.5$: t versus $C^{0.5}$; $\lambda = 0.5$
- $n = 1$: t versus $\ln C$; $\lambda = 0$
- $n = 1.5$: t versus $C^{-0.5}$; $\lambda = -0.5$
- $n = 2$: t versus C^{-1}; $\lambda = -1$

In summary, comparison of the equations establishes the relationship:

$$\lambda = 1 - n$$

2.2 Applying the Box–Cox Method to Time–Concentration Data

To examine the application of the Box–Cox method to problem-solving, three classic time–concentration data problems were selected to test its robustness. Each set of experimental t–C data was analyzed using the following standardized statistical procedure.

First, an initial linear model of concentration against time, $C = f(t)$, was constructed without mathematically transforming the original concentration data. Although this model does not necessarily have kinetic significance, it serves as a baseline for the Box–Cox transformation and for assessing the distribution of the original observations.

Second, maximum likelihood estimation (MLE) was used to calculate the log-likelihood for different values of λ based on the baseline model. The peak of the likelihood curve was used to identify the optimal transformation parameter, producing the greatest normality of the model residuals and the strongest linearity.

Third, the original concentration data were transformed using the power suggested by the optimal λ, and ordinary least squares (OLS) regression was performed on the transformed variable against time t.

Finally, regression statistics were used to assess the transformed model.

2.2.1 Second-Order Reaction: Diels–Alder Dimerization of 2,5-Dimethyl-3,4-diphenylcyclopentadienone

This is a typical Diels–Alder [4+2] cycloaddition reaction: two cyclopentadienone monomers act as the diene and dienophile, respectively, and combine to form a stable dimer. The system is particularly easy

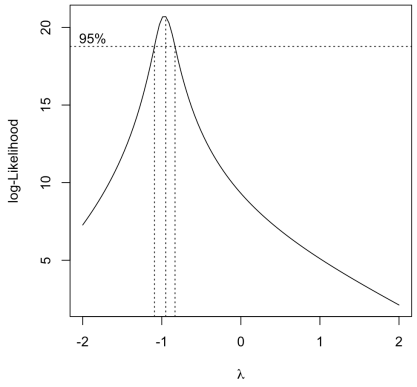
to observe in a physical chemistry experiment. The monomer is deep red because of its highly conjugated electronic structure, whereas the resulting dimer is colorless. Spectrophotometry is therefore commonly used to monitor the change in absorbance—which is proportional to concentration—over time. [5]

Table 14.4.2: Rates of Reaction as a Function of Monomer Concentration for an Initial Monomer Concentration of 0.0054 M

Time (min)	[Monomer] (M)	Instantaneous Rate (M/min)
10	0.0044	8.0×10^{-5}
26	0.0034	5.0×10^{-5}
44	0.0027	3.1×10^{-5}
70	0.0020	1.8×10^{-5}
120	0.0014	8.0×10^{-6}

Figure 1: Time–concentration data for the second-order reaction

In theory, because the reaction involves the collision and combination of two identical monomer molecules, its elementary step follows second-order kinetics. The observed time–concentration data were subjected to the Box–Cox transformation to find the optimal λ and determine whether the statistical result agreed with the expected second-order behavior, corresponding to $\lambda = -1$. [3]




```
Call:
lm(formula = 1/C ~ libretext$t)

Residuals:
      1       2       3       4       5       6
 3.2851  0.9345 -3.3217 -7.0578  7.0326 -0.8727

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 181.90008    3.37844   53.84 7.12e-07 ***
libretext$t   4.44382    0.05578   79.67 1.49e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.539 on 4 degrees of freedom
Multiple R-squared:  0.9994,    Adjusted R-squared:  0.9992
F-statistic:  6347 on 1 and 4 DF,  p-value: 1.488e-07
```

Figure 2: Box–Cox transformation and linear regression results for the second-order reaction

As shown in Figure 2, left, the likelihood curve reaches its maximum near $\lambda = -1$, and the 95% confidence interval includes -1. According to the relationship derived above, this strongly suggests that the reaction follows second-order kinetics, associated with a reciprocal transformation. Figure 2, right, shows that the transformed regression model has an R^2 of 0.9994, a very small residual standard deviation, and a p-value much smaller than 0.05.

2.2.2 First-Order Reaction: Decomposition of Hydrogen Peroxide

The second case is the classic catalytic decomposition of hydrogen peroxide (H_2O_2). In a typical physical chemistry experiment, a catalyst such as KI or catalase is added under constant-temperature conditions, and

the remaining H_2O_2 concentration is measured periodically by titration, such as potassium permanganate titration, or by measuring the volume of gas produced. [4]

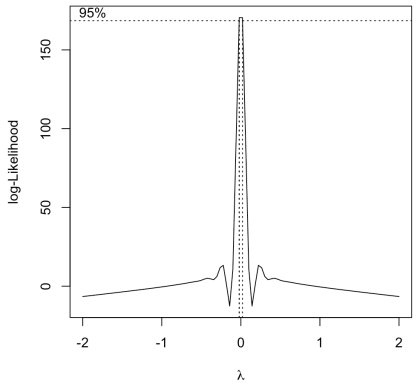
Solution

The data from Figure 12.2 are tabulated below, and a plot of $\ln[H_2O_2]$ is shown in Figure 12.9.

Time (h)	$[H_2O_2]$ (M)	$\ln[H_2O_2]$
0.00	1.000	0.000
6.00	0.500	–0.693
12.00	0.250	–1.386
18.00	0.125	–2.079
24.00	0.0625	–2.772

Figure 3: Time–concentration data for the first-order reaction

From a mechanistic perspective, under suitable conditions and in the presence of particular catalysts, hydrogen peroxide decomposition commonly exhibits first-order behavior. This means that the reaction rate is proportional to the first power of the instantaneous H_2O_2 concentration. According to the derivation above, a first-order reaction corresponds to the Box–Cox parameter $\lambda = 0$, or the logarithmic transformation $\ln C$. The recorded hydrogen peroxide concentration–time data were analyzed using the Box–Cox transformation.




```
Call:
lm(formula = log(C) ~ Openstax$t)

Residuals:
         1          2          3          4          5
-6.160e-17  3.232e-17  2.359e-17  1.023e-16 -9.659e-17

Coefficients:
              Estimate Std. Error    t value Pr(>|t|)
(Intercept) -1.199e-16  7.097e-17 -1.689e+00     0.19
Openstax$t  -1.155e-01  4.829e-18 -2.392e+16   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.163e-17 on 3 degrees of freedom
Multiple R-squared:      1,     Adjusted R-squared:      1
F-statistic: 5.723e+32 on 1 and 3 DF,  p-value: < 2.2e-16
```

Figure 4: Box–Cox transformation and linear regression results for the first-order reaction

Similarly, Figure 4, left, shows that the likelihood curve reaches its maximum near $\lambda = 0$, and the 95% confidence interval includes 0, indicating first-order kinetics. The regression results in Figure 4, right, show an R^2 of 1 for the transformed model and a p-value much smaller than 0.05.

2.2.3 Reaction of Order 1.5: An Example from *Concise Physical Chemistry*

The final case is a textbook example intended for the half-life method. [1] The problem does not specify the chemical reaction involved, but its time–concentration data are well suited to analysis by the half-life

method.

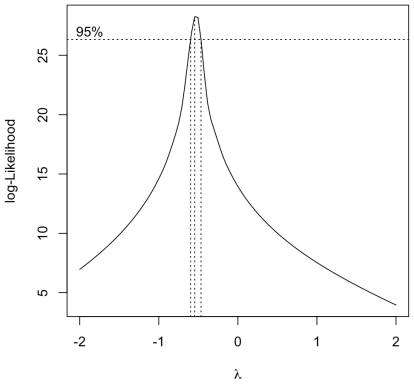
例 8-2 298K 时测得溶液中某分解反应 2A ⟶ B+C 的如下数据：

t/s	0	12	29	41	58	83
c_A/(mol • m^{-3})	4	3.2	2.4	2.0	1.6	1.2

试求该反应的级数。

Figure 5: Time–concentration data for the reaction of order 1.5

The concentration–time data are analyzed using the Box–Cox transformation to demonstrate that the correct conclusion can be reached even without recognizing that the half-life method is applicable.




```
Call:
lm(formula = reciprocal(sqrt(Ca..mol.m3.)) ~ T.s)

Residuals:
         1          2          3          4          5          6
-0.0004674 -0.0013028  0.0003865  0.0021437  0.0008155 -0.0015755

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 5.005e-01  1.068e-03   468.7 1.24e-10 ***
T.s         4.988e-03  2.301e-05   216.8 2.72e-09 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.001566 on 4 degrees of freedom
Multiple R-squared:  0.9999,    Adjusted R-squared:  0.9999
F-statistic: 4.698e+04 on 1 and 4 DF,  p-value: 2.718e-09
```

Figure 6: Box–Cox transformation and linear regression results for the reaction of order 1.5

As shown in Figure 6, left, the likelihood curve reaches its maximum near $\lambda = -0.5$, and its 95% confidence interval includes 0, indicating kinetics of order 1.5. The regression results in Figure 6, right, show an R^2 of 0.9999 for the transformed model and a p-value much smaller than 0.05. These results show an excellent linear relationship among the data points after the transformation suggested by the Box–Cox method, thereby rigorously validating the assigned reaction order.

The example of the 1.5-order reaction is particularly useful for illustrating the method's strength. If students do not recognize that the half-life method can be used and instead apply trial-and-error linear regression, they may easily conclude that the reaction is second order. Indeed, a tutor who helped the author's class prepare for the physical chemistry final examination made this error. Regressing the data after the reciprocal transformation associated with a second-order reaction can also produce a high R^2. Even so, the 95% confidence interval for λ in the Box–Cox analysis does not include -1, indicating that the method can distinguish between reactions of order 1.5 and order 2.

3. Summary and Discussion

This paper has examined the use of the Box–Cox transformation to determine chemical reaction order through theoretical derivation and several case studies. The main findings and discussion are as follows.

3.1 Theoretical Relationship

The derivation establishes a definite mathematical relationship between the Box–Cox transformation parameter λ and reaction order n:

$$\lambda = 1 - n$$

The transformation provides a unified way to convert integrated kinetic equations into linear form, covering zero-order ($\lambda = 1$), first-order ($\lambda = 0$), second-order ($\lambda = -1$), and half-integer-order reactions.

3.2 Practical Validation

Analysis of the Diels–Alder dimerization, hydrogen peroxide decomposition, and textbook reaction of order 1.5 demonstrates that the optimal λ estimated by MLE and its 95% confidence interval can be used to determine reaction order accurately. Even when the data are highly similar, the method can effectively distinguish between reactions of order 1.5 and order 2, avoiding biases that may arise from subjective judgment.

3.3 Limitations and Future Work

Nevertheless, this study has substantial limitations. The time–concentration data used here come primarily from classic physical chemistry textbooks and examination problems. Such data are often idealized, with very high signal-to-noise ratios, and sampling points are frequently distributed around characteristic points of the reaction.

During the investigation, the author found that obtaining raw time–concentration trajectories from real research settings is extremely challenging: most research papers tend to report calculated rate constants directly rather than the original observations. Future work could therefore apply the statistical model to more complex real-world research data, such as complex catalytic reactions involving multiple intermediate states or systems with substantial experimental noise.

Testing the method in real research settings would allow further assessment of whether new identification biases or algorithmic limitations arise when it is applied to nonideal data, helping to refine and improve this statistical framework for kinetic analysis.

4. Conclusions

This study establishes the mathematical relationship $\lambda = 1-n$ between the Box–Cox parameter and reaction order, providing a unified linearization framework for determining a range of reaction orders, including half-integer orders. In case studies including Diels–Alder dimerization and hydrogen peroxide decomposition, the method showed high analytical accuracy and identified complex kinetic behavior such as a reaction order of 1.5. Future research should address the limitations of the currently used idealized textbook data by testing the method on complex data from real research settings, thereby further validating and improving the statistical model.

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Supporting Information

All data and RStudio code are summarized in the "Supplementary Information" ZIP file.

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